

Performance Testing Phase

Project Performance Test

Project Title	India's Agricultural Crop Production Analysis
Domain	Agriculture / Data Analytics & Business Intelligence
Tool	Tableau Desktop / Tableau Public
Dataset	Historical Indian Crop Production Data, 1997–2021
Prepared By	Divya Kauluri (Team Project — completed individually; the team lead and other assigned members did not contribute to the work)
Mentor	N/A — Individual Submission
GitHub Repository	https://github.com/divyakauluri-spec/India-s-agricultural-crop-production-analysis
Tableau Public Link	https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject
Date	02/07/2026

Project Context

This report presents a comprehensive visual exploration of India's agricultural cultivation using 25 years (1997–2021) of historical crop production data. Nine Tableau visualizations — a KPI Summary panel, State-wise Production Map, Year-wise Total Production chart, Trend of Agricultural Yield chart, Cultivated Area by Crop chart, State-wise Cultivated Area Comparison chart, Leading States by Production chart, Production Distribution by Unit Type donut, and Season-wise Production pie — are combined into four role-specific dashboards and a five-point guided Story, enabling stakeholders to uncover patterns, identify areas of growth or concern, and make data-driven decisions.

By harnessing the power of Tableau, this report presents the data in a visually appealing, interactive form that lets readers explore the intricacies of India's agricultural cultivation for themselves.

Scenario 1: Small-Scale Farmer

Rajesh is a small-scale farmer in Uttar Pradesh who wants to maximize his crop yield and income. However, he struggles to decide which crops to grow each season due to changing weather patterns and market demand. By analyzing historical agricultural data (1997–2021), Rajesh needs insights into seasonal crop trends, regional productivity, and high-performing crops to make better farming decisions and reduce risk. The Farmer's View dashboard serves Rajesh through the Cultivated Area by Crop and Season-wise Production charts.

Scenario 2: Government Policy Analyst

Ananya works with the Ministry of Agriculture and is responsible for designing policies to improve agricultural productivity across India. She needs to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations. Using Tableau dashboards, she aims to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively. The Policy Analyst's View dashboard serves Ananya through the State-wise Production Map and Trend of Agricultural Yield chart.

Scenario 3: Agribusiness Investor

Vikram is an agribusiness investor looking to invest in high-growth agricultural sectors. He needs to analyze historical crop production data to identify profitable crops, emerging trends, and regions with consistent growth. By leveraging visual analytics in Tableau, Vikram seeks to minimize investment risk and make informed decisions on where and

what to invest in the agricultural market. The Investor’s View dashboard serves Vikram through the Leading States by Production and Cultivated Area by Crop charts.

5.1 Performance Testing Results

Test #	Description	Expected Result	Actual Result	Status
PT-01	Amount of Data Rendered	All rows render without error	345,407 rows loaded, 12 columns — no errors	PASS
PT-02	Season Filter Response	Map updates within 3 seconds	Map redrew fully in ~1.1 seconds	PASS
PT-03	Crop Year Slider Response	Chart updates within 3 seconds	Line chart redrew in ~0.9 seconds	PASS
PT-04	Calculated Fields	No null or error values	Yield and Crop_Year_Numeric compute correctly	PASS
PT-05	Number of Visualizations	9 charts + 4 dashboards + 1 story	Confirmed: 9 chart sheets, 4 dashboards, 1 story	PASS
PT-06	Cross-Browser Compatibility	Loads and filters work in all browsers	Verified in Chrome, Edge, and Safari — no errors	PASS
PT-07	Tooltip Accuracy	Correct values shown on hover	Verified across all chart types	PASS
PT-08	Story Navigation	Smooth transition between story points	All story points navigate correctly with captions	PASS

5.2 Performance Summary

Metric	Value
Total Dataset Rows	345,407
Total Columns	12
Number of Calculated Fields	3 (Yield, Crop_Year_Numeric, Production_Normalized)
Number of Filters Used	3 (Season, Crop Year Numeric, Crop Top-N)
Number of Visualizations / Graphs	9
Number of Dashboards	4
Number of Story Points	5
Overall Performance Rating	PASS — No performance issues detected